connecting XB and BD. The pattern looks like two wings. When trying to visualize this, consumption of alcohol may be a plus.

As I mentioned, the pattern's turns are found using Fibonacci numbers. Let's go through this pattern to see how the bat in Figure 4.2 qualifies.

BA/XA retrace. The ratio of leg BA to XA should be either .382 or .5. Here are the prices used in the retrace or extension calculations for the bearish bat. The high price of point X is 56.93. The low at point A is 47.81, B peaks at 52.00, C bottoms at 49.71, and the pattern ends at D with a high price of 55.98.

To determine the retrace, I use the height (high–low range) of the price bar at the target to determine whether or not it spans a Fibonacci number. For example, the BA/XA turn using the high price of B would be (52.00 – 47.81)/(56.93 – 47.81) or .46. Using the low price at B (50.90) in the equation gives (50.90 – 47.81)/(56.93 – 47.81) or .34. The range .34 to .46 straddles the .382 Fibonacci number, so I allow it as a valid XAB turn.

Using this method gives a lot of leeway to the pattern's turns but even so, the pattern is rare. Trying to narrow the turn window would limit even more patterns from being found, so this is the algorithm I chose. The software you use may think differently.

BC/BA retrace. In a manner similar to the BA/XA retrace, I plugged in the numbers for the BC/BA retrace (using the low at C). They are (52 – 49.71)/(52 – 47.81) or .55. Using the high at C (51.65) gives (52 – 51.65)/(52 – 47.81) or .08. The range of .08 to .55 encompasses .382 and .5, so I allow the turn as valid.

DC/BC extension. In a similar manner, I find the DC extension of BC. That's (55.98 – 49.71)/(52 – 49.71) or 2.74. Using the low at D (54), we get an extension of 1.87. The 1.87 to 2.74 values straddle all of the listed numbers (any one of which will suffice) except 1.618, so it qualifies as a valid extension.

DA/XA retrace. Supposedly, this ratio is critical, so I allow point D to miss the number by 3% (.03). I do *not* use the high–low range of D in the equation and I use 3% instead of 1% because this pattern is rare enough as it is. We have (55.98 – 47.81)/(56.93 – 56.93) or .9. The .9 value is near the .886 target, so the pattern qualifies as a valid bearish bat.